Wigner Resolution from Action Capacity

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Smoothing the Wigner function gives a non-negative phase-space portrait of a quantum state, with the smoothing scale conventionally left free. We show that the scale is fixed by the state. The state's covariance defines a Heisenberg cell of action capacity $\pi\,\Delta x\,\Delta p$, admitting a family of de Gosson quantum blobs deforming at constant action $h/2$. Convolving the Wigner function with the kernel of a single $h/2$ blob renders it non-negative. Integrating over the family resolves the portrait at $\pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the Heisenberg cell. The portrait is everywhere non-negative and carries the structure of the Wigner function undisplaced. Wigner resolution carries no free parameter, set by the state's action capacity alone and sharpening as that capacity grows.

The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise below Planck's quantum of action, where Hudson's theorem forces it negative [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6]. A non-negative distribution supports classical sampling [7–11] and follows from smoothing the Wigner function at a freely chosen scale. Here the state's action capacity $\pi\,\Delta x\,\Delta p$ sets that scale without a free parameter.

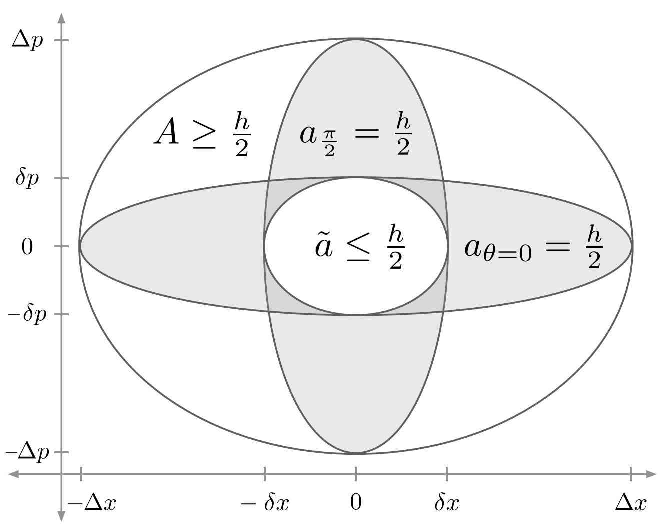

FIG. 1. **Heisenberg cell, quantum blobs at principal axes, and quorum cell.** The Heisenberg cell is the state's covariance ellipsoid with action capacity $A = \pi\,\Delta x\,\Delta p \geq h/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The quorum cell $\tilde{a}$ is an ellipse with semi-axes $\delta x, \delta p$ and action $\tilde{a} = \pi\,\delta x\,\delta p = \pi\hbar^2/(\Delta x\,\Delta p)$. A circumscribes $a_{\theta=0}$ and $a_{\pi/2}$, which circumscribe $\tilde{a}$.

Planck divided phase space into elementary regions of action [12]. Heisenberg's uncertainty principle $\sigma_x\,\sigma_p \geq \hbar/2$ [13–15] fixes their minimum area. Geometrically, de Gosson gave this action region the form of the state's covariance ellipse, with symplectic capacity $\pi\,\Delta x\,\Delta p$ [16]

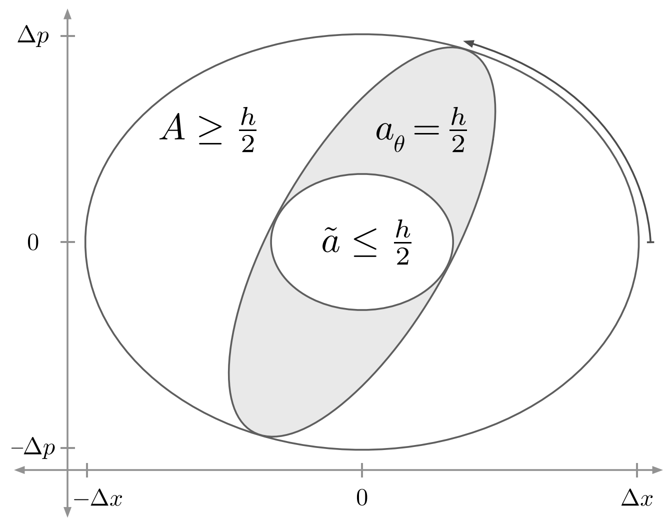

FIG. 2. **Heisenberg cell, quadrature family of quantum blobs, and quorum cell.** The Heisenberg cell A circumscribes the quantum blob family $\{a_\theta\}$, which circumscribes the quorum cell $\tilde{a}$. Across θ the family deforms at constant action $a_\theta = h/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $a_{\theta=0}$ and $a_{\pi/2}$ from Fig. 1.

and semi-axes $\Delta x = \sqrt{2}\,\sigma_x$ and $\Delta p = \sqrt{2}\,\sigma_p$. This is the Heisenberg cell (Figs. 1, 2) of action capacity,

$$A = \pi\,\Delta x\,\Delta p \geq h/2. \tag{1}$$

The Wigner function carries structure at small scales. Zurek located those scales, $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$, from the state's action capacity [17]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2, \qquad a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2. \tag{2}$$

The capacities $\Delta x, \Delta p$ run along the major axes, the resolutions $\delta x, \delta p$ along the minor.

Between these principal members the family $\{a_\theta\}$ deforms across $\theta \in [0, \pi)$ at constant action $h/2$ (Fig. 2). The Heisenberg cell A circumscribes each a_θ. In the

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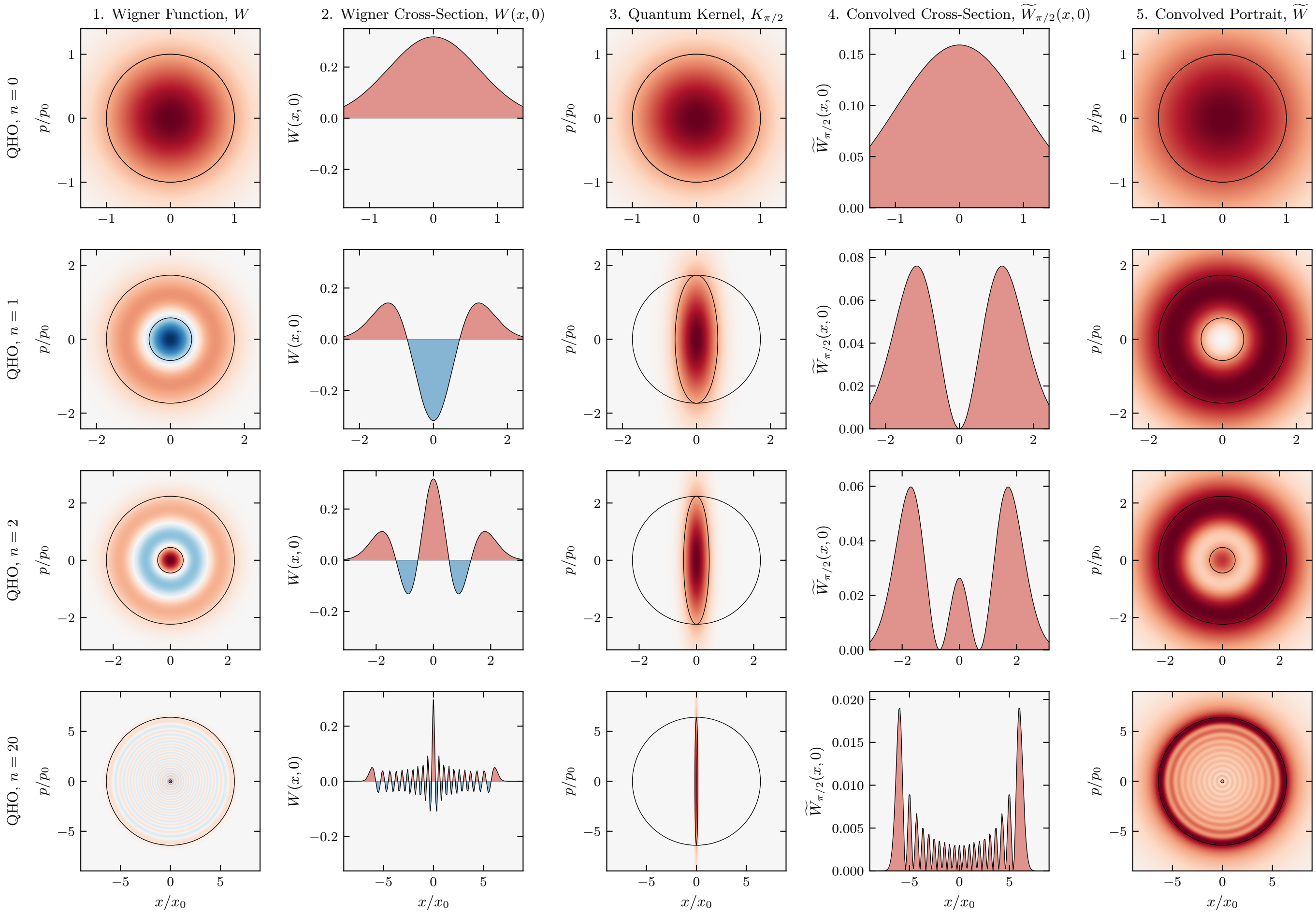

FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(h/2) = 2n+1$. Column 1: Wigner function $W(x,p)$ (negativity in blue) overlaid with the Heisenberg cell A and quorum cell $\tilde{a}$. Centered at the covariance origin, $\tilde{a}$ marks the resolution of the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x,0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $a_{\pi/2}$ (action $h/2$, restoring Planck's quantum of action). Column 4: Convolved cross-section $\widetilde{W}_{\pi/2}(x,0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\widetilde{W}(x,p)$, integrated over the blob family across all θ, overlaid with A and $\tilde{a}$ (symplectic polar duals). It is everywhere non-negative, with positive lobes at the positions of those in W, resolved at $\tilde{a}$. For the vacuum ($n = 0$), A and $\tilde{a}$ coincide as a Gaussian blob at $A = h/2$. As n increases, A grows and $\tilde{a}$ reciprocally shrinks ($\tilde{a}\,A = (h/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving $\tilde{a}$ as a fine central cell. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$ it is one ellipse turned through θ,

$$a_\theta = \left\{(x,p) : X_\theta^2 + \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2 \le 1\right\}, \tag{3}$$

inscribed in the unit disk. The quadratures $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$ are those of homodyne tomography at phase θ.

The Wigner function of the state is

$$W(x,p) = \frac{1}{\pi\hbar}\int \psi^*(x+s)\,\psi(x-s)\,e^{2ips/\hbar}\,ds. \tag{4}$$

The Wigner function of each quantum blob a_θ is the centered Gaussian whose $h/2$ ellipse is a_θ,

$$K_\theta(x,p) = \frac{1}{\pi\hbar}\exp\Big[-X_\theta^2 - \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2\Big]. \tag{5}$$

The convolution of W with each quantum kernel K_θ is

$$\widetilde{W}_\theta(x,p) = (W * K_\theta)(x,p). \tag{6}$$

Convolving W with the kernel K_θ of a single quantum blob a_θ lifts it up to Planck's quantum of action and yields a non-negative function [8, 9, 18].

Zurek also located the scale $a \sim \hbar \times (\hbar/A)$ [17, 19] of the chessboard structure in the compass state. We identify its rotationally invariant form as the quorum cell $\tilde{a}$

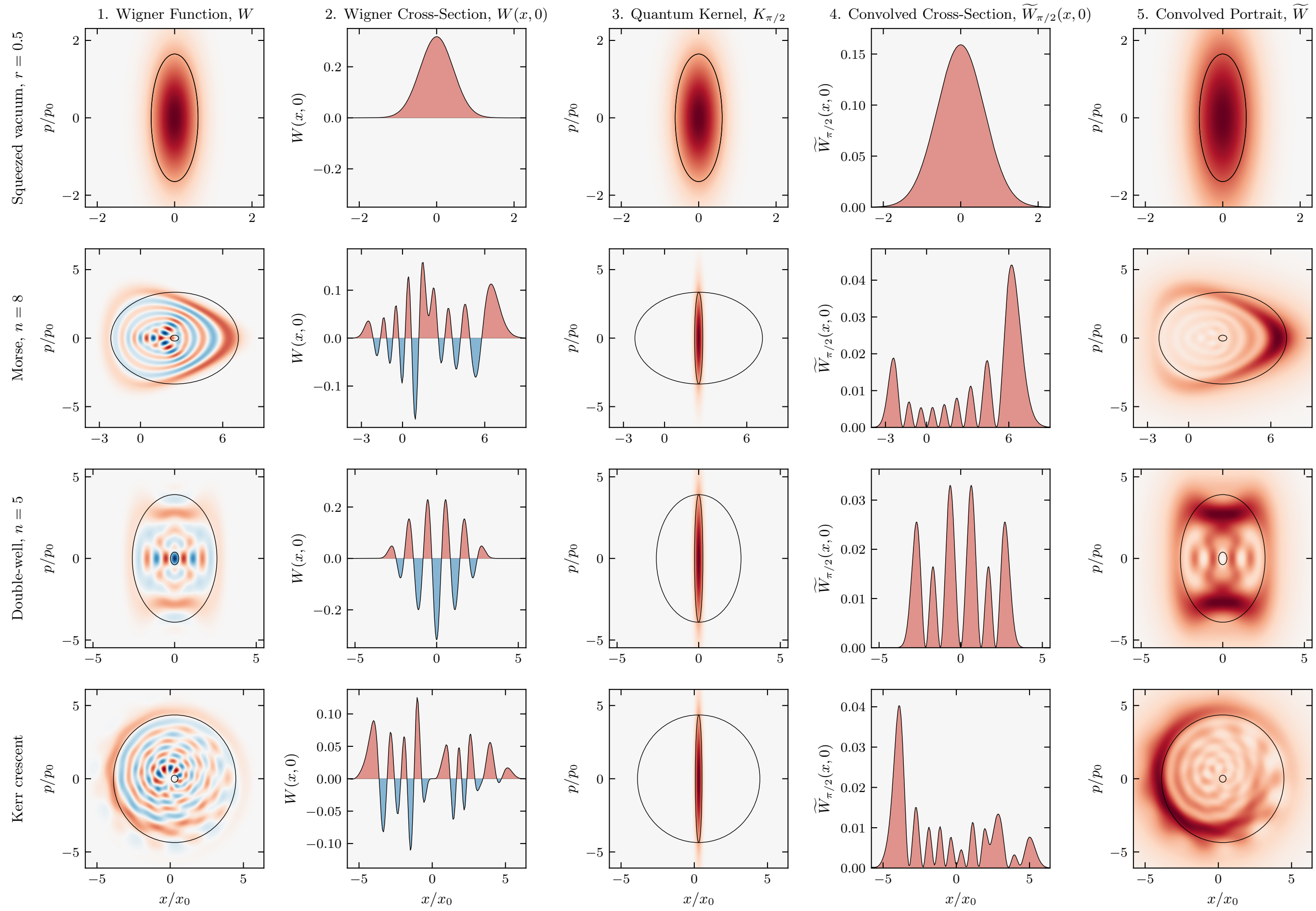

FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the cell overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at $\tilde{a}$.

(Fig. 5, column 1). Named after the quorum of observables in optical tomography [20], $\tilde{a}$ is the ellipse with semi-axes $\delta x, \delta p$ (Figs. 1, 2) and action

$$\tilde{a} = \pi\, \delta x\, \delta p = \frac{\pi \hbar^2}{\Delta x\, \Delta p}. \tag{7}$$

It is the resolution of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1) and the symplectic polar dual of the Heisenberg cell A [21, 22]. The duality is reflexive, so A and $\tilde{a}$ each fix the other, and antimonotone, so $\tilde{a}$ shrinks as A grows. Each blob a_θ is circumscribed by A and circumscribes $\tilde{a}$, nesting as $A \supseteq a_\theta \supseteq \tilde{a}$ (Fig. 2). Resolution and capacity are reciprocal at the quantum of action, $\tilde{a}\, A = (h/2)^2$.

A single blob commits to a quadrature the state does not favor. Integrating $\widetilde{W}_\theta$ over the family favors none, yielding the phase-space portrait,

$$\widetilde{W}(x,p) = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta(x,p)\, d\theta. \tag{8}$$

Each $\widetilde{W}_\theta \geq 0$, so $\widetilde{W} \geq 0$ everywhere. Where standard tomography inverts the marginals to recover the Wigner quasi-probability W [23–25], convolving W with $\{K_\theta\}$ over the same θ builds the non-negative portrait $\widetilde{W}$.

The convolution and the integral are linear, so the family reduces to a single convolution,

$$\widetilde{W} = W * K, \qquad K = \frac{1}{\pi} \int_0^\pi K_\theta\, d\theta, \tag{9}$$

and the breadth of the kernel's Fourier transform $\widehat{K}$ sets the portrait's resolution. With (ξ_x, ξ_p) the chord conjugate to (x,p), integrating over θ bounds $\widehat{K}$ above by a single Gaussian,

$$\widehat{K}(\xi_x, \xi_p) \leq \exp\left[-\tfrac{1}{4}\left(\delta x^2\, \xi_x^2 + \delta p^2\, \xi_p^2\right)\right], \tag{10}$$

with equality at the origin. The bound cuts off at δx and δp, so the portrait carries no structure finer than $\pi\, \delta x\, \delta p = \tilde{a}$, derived exactly in the Supplemental Material [26].

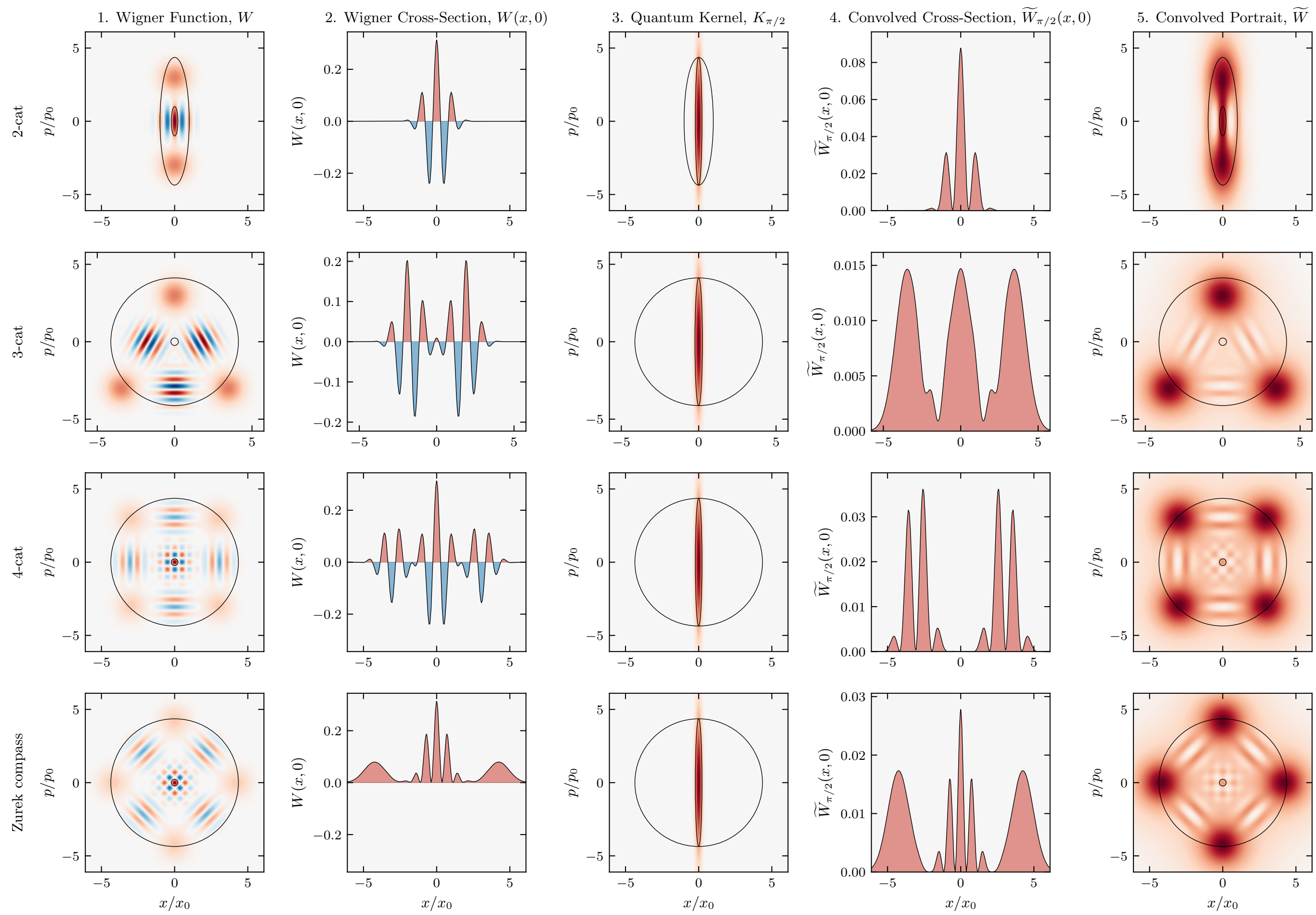

FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\widetilde{W}$ (Column 5) is everywhere non-negative, with positive lobes at the positions of those in W, resolved at $\tilde{a}$. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

A quantum blob a_θ too large for A to circumscribe, or too small to circumscribe $\tilde{a}$, is a blob of another state and portrays that state, not this one [21, 27]. Every $h/2$ blob of this state circumscribes the quorum cell $\tilde{a}$, so no $h/2$ smoothing of this state resolves finer, and a kernel below $h/2$ fails to render the portrait non-negative [8, 9]. The finest resolution is $\tilde{a}$, reached by the family $\{a_\theta\}$ across $\theta \in [0, \pi)$.

A fixed vacuum-width Gaussian [7] smooths every state by the same amount, matching only the vacuum $A = h/2$ and erasing the fine structure of larger states $A > h/2$. Methods with a free parameter [28–30] resolve more finely, the scale set from outside the state. Here the state sets its own scale. Its action capacity A fixes the quorum cell $\tilde{a}$ as the symplectic polar dual, with nothing left free.

The principal frame, where the Heisenberg cell A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [18], so the map that diagonalizes a state's covariance carries it to this frame at preserved action. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely, and the sub-Planck structure is already present for states with $A > h/2$. In higher dimensions, action $h/2$ no longer singles out a blob [27], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

Figures 3, 4, and 5 show numerical results across twelve states. The portrait is non-negative to machine precision, the chord-domain transfer carrying W to $\widetilde{W}$ matches prediction with no free parameter, and the cross-spectral phase between W and $\widetilde{W}$ vanishes, so the structure is resolved at $\tilde{a}$ without displacement [26].

The uncertainty product $\sigma_x\sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [31] and Bohr [32] identified as requiring special care. The quantum blob does not. As A grows, the family $\{a_\theta\}$ deforms at fixed action $h/2$. Squeezing reaches the same resolution by external apparatus [33]. The scale called sub-Planck [17] is the minor axis δ of an $h/2$ blob, narrowing as the conjugate capacity Δ grows. A Schrödinger cat with capacities

$\Delta x \sim 10$ cm and $\Delta p \sim 1$ kg·m/s carries structure in its Wigner function at resolution $\delta x = \hbar/\Delta p \sim 10^{-34}$ m [34]. The correspondence principle follows from $A \to \infty$, the physical limit underlying the heuristic $h \to 0$.

As the action capacity grows to $\pi\,\Delta x\,\Delta p \geq h/2$ the resolution sharpens to $\pi\hbar^2/(\Delta x\,\Delta p) \leq h/2$. We read this structure as the geometry of phase space itself, which de Gosson takes as prior to the uncertainty principle [35].

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[1] E. Wigner, On the quantum correction for thermodynamic equilibrium, Physical Review **40**, 749 (1932).

[2] M. Hillery, R. F. O'Connell, M. O. Scully, and E. P. Wigner, Distribution functions in physics: Fundamentals, Physics Reports **106**, 121 (1984).

[3] R. L. Hudson, When is the Wigner quasi-probability density non-negative?, Reports on Mathematical Physics **6**, 249 (1974).

[4] V. Veitch, C. Ferrie, D. Gross, and J. Emerson, Negative quasi-probability as a resource for quantum computation, New Journal of Physics **14**, 113011 (2012).

[5] A. Mari and J. Eisert, Positive Wigner functions render classical simulation of quantum computation efficient, Physical Review Letters **109**, 230503 (2012).

[6] A. Kenfack and K. Życzkowski, Negativity of the Wigner function as an indicator of non-classicality, Journal of Optics B: Quantum and Semiclassical Optics **6**, 396 (2004).

[7] K. Husimi, Some formal properties of the density matrix, Proc. Phys.-Math. Soc. Jpn., 3rd Ser. **22**, 264 (1940).

[8] N. D. Cartwright, A non-negative Wigner-type distribution, Physica A: Statistical Mechanics and its Applications **83**, 210 (1976).

[9] M. de Gosson, Cellules quantiques symplectiques et fonctions de Husimi–Wigner, Bulletin des Sciences Mathématiques **129**, 211 (2005).

[10] D. Dragoman, Phase space formulation of quantum mechanics. Insight into the measurement problem, Phys. Scr. **72**, 290 (2005).

[11] A. Polkovnikov, Phase space representation of quantum dynamics, Annals of Physics **325**, 1790 (2010).

[12] M. Planck, *Vorlesungen über die Theorie der Wärmestrahlung* (Johann Ambrosius Barth, Leipzig, 1906) english translation: M. Masius, *The Theory of Heat Radiation* (Blakiston, Philadelphia, 1914).

[13] E. H. Kennard, Zur Quantenmechanik einfacher Bewegungstypen, Zeitschrift für Physik **44**, 326 (1927).

[14] H. P. Robertson, The uncertainty principle, Physical Review **34**, 163 (1929).

[15] E. Schrödinger, Zum Heisenbergschen Unschärfeprinzip, Sitzungsber. Preuss. Akad. Wiss., Phys.-Math. Kl. , 296 (1930).

[16] M. A. de Gosson, The symplectic camel and the uncertainty principle: The tip of an iceberg?, Foundations of Physics **39**, 194 (2009).

[17] W. H. Zurek, Sub-Planck structure in phase space and its relevance for quantum decoherence, Nature **412**, 712 (2001).

[18] E. Cordero, M. de Gosson, M. Dörfler, and F. Nicola, On the symplectic covariance and interferences of time-frequency distributions, SIAM Journal on Mathematical Analysis **50**, 2178 (2018).

[19] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Reviews of Modern Physics **75**, 715 (2003).

[20] G. M. D'Ariano, L. Maccone, and M. G. A. Paris, Quorum of observables for universal quantum estimation, Journal of Physics A: Mathematical and General **34**, 93 (2001).

[21] M. de Gosson and C. de Gosson, Symplectic polar duality, quantum blobs, and generalized Gaussians, Symmetry **14**, 1890 (2022).

[22] M. A. de Gosson, Polar duality and the reconstruction of quantum covariance matrices from partial data, Journal of Physics A: Mathematical and Theoretical **57**, 205303 (2024).

[23] K. Vogel and H. Risken, Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase, Physical Review A **40**, 2847 (1989).

[24] A. I. Lvovsky and M. G. Raymer, Continuous-variable optical quantum-state tomography, Reviews of Modern Physics **81**, 299 (2009).

[25] M. A. de Gosson, Symplectic Radon transform and the metaplectic representation, Entropy **24**, 761 (2022).

[26] B. S. Mulloy, Code, figures, and supplemental material for "Wigner Resolution from Action Capacity" (2026), the supplemental material is archived in this deposit, under the same DOI.

[27] M. A. de Gosson, Quantum blobs, Foundations of Physics **43**, 440 (2013).

[28] K. Wódkiewicz, Operational approach to phase-space measurements in quantum mechanics, Phys. Rev. Lett. **52**, 1064 (1984).

[29] D. M. Appleby, Generalized Husimi functions: Analyticity and information content, J. Mod. Opt. **46**, 825 (1999).

[30] K. E. Cahill and R. J. Glauber, Density operators and quasiprobability distributions, Physical Review **177**, 1882 (1969).

[31] W. Heisenberg, The physical content of quantum kinematics and mechanics, in *Quantum Theory and Measurement*, edited by J. A. Wheeler and W. H. Zurek (Princeton University Press, 1983) pp. 62–84.

[32] N. Bohr, The quantum postulate and the recent development of atomic theory, Nature **121**, 580 (1928).

[33] D. F. Walls, Squeezed states of light, Nature **306**, 141 (1983).

[34] E. Schrödinger, The present situation in quantum mechanics, Proceedings of the American Philosophical Society **124**, 323 (1980), translated by J. D. Trimmer from *Naturwissenschaften* **23** (1935) 807–812, 823–828, 844–849.

[35] M. A. de Gosson, On quantum indeterminacy and the uncertainty principle (2026), arXiv:2605.01103 [quant-ph].

[36] N. Lambert *et al.*, QuTiP 5: The quantum toolbox in Python, Phys. Rep. **1153**, 1 (2025).

[37] P. Virtanen *et al.*, SciPy 1.0: Fundamental algorithms for scientific computing in Python, Nat. Methods **17**, 261 (2020).